X&X Ceramics — To'liq Texnik Topshiriq (TZ)

Versiya: 1.0
Sana: May 2026
Loyiha: X&X Ceramics korporativ websayti + Admin Telegram Bot
Buyurtmachi: Xasanboy & Xusanboy (X&X Ceramics)

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1. Loyiha haqida umumiy ma'lumot

1.1 Maqsad

X&X Ceramics — egizak ustalar Xasanboy va Xusanboy tomonidan boshqariladigan hunarmandchilik korxonasi uchun professional korporativ websayt va admin boshqaruv tizimini yaratish. Websayt potentsial mijozlarga korxona mahsulotlari va xizmatlari haqida ma'lumot berishi, ishonch hosil qilishi va murojaat imkonini yaratishi kerak.

1.2 Nom to'g'risida taklif

X&X Ceramics nomi aniq va to'g'ri. Qo'shimcha variantlar:

Variant	Izoh
X&X Ceramics ✅	Egizaklar ramzi, lakonik va esda qoladigan
Twin Craft	"Egizak hunarmandlar" ma'nosi, xalqaro talaffuzi qulay
XH Studio	Xasanboy + Xusanboy bosh harflari, studio imidzhi
Ikki Usta	O'zbekcha versiya, mahalliy bozor uchun qulay

Tavsiya: X&X Ceramics eng to'g'ri tanlov — qisqa, brend sifatida registratsiya qilish oson, xalqaro bozorlarda ham tushunarli.

1.3 Loyiha tarkibi

- **Frontend:** Korporativ websayt (React / Next.js)
- **Backend:** REST API (Node.js / Express)
- **Ma'lumotlar bazasi:** PostgreSQL
- **Admin panel:** Telegram bot
- **Fayl saqlash:** Cloudinary yoki AWS S3

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2. Texnik talablar

2.1 Frontend texnologiya stegi

Qatlam	Texnologiya	Izoh
Framework	Next.js 14 (App Router)	SEO uchun SSR/SSG qo'llab-quvvatlash
UI kutubxona	Tailwind CSS + shadcn/ui	Tezkor va responsive dizayn
Animatsiyalar	Framer Motion	Silliq sahifa o'tishlari
Rasmlar	next/image + Cloudinary	Optimizatsiya va CDN
Formalar	React Hook Form + Zod	Validatsiya
Holat boshqaruvi	Zustand yoki React Query	API ma'lumotlari uchun
TypeScript	Ha	Xavfsiz kod yozish

2.2 Backend texnologiya stegi

Qatlam	Texnologiya	Izoh
Runtime	Node.js 20 LTS	Barqaror versiya
Framework	Express.js	REST API
ORM	Prisma	PostgreSQL bilan ishlash
Auth	JWT + bcrypt	Admin autentifikatsiya
Fayl yuklash	Multer + Cloudinary SDK	Rasm va video yuklash
Validatsiya	Joi / Zod	So'rovlarni tekshirish
Logging	Winston	Xatolarni kuzatish

2.3 Muhit (Environment)

```
Development: localhost
Staging: staging.xxceramics.uz
Production: xxceramics.uz (yoki xxceramics.com)
```

3. Websayt strukturasi

3.1 Sahifalar xaritasi (Sitemap)

```
/ (Bosh sahifa)
├── /portfolio (Ishlar portfeli)
│   ├── /portfolio/lepka
│   ├── /portfolio/tyaga
│   ├── /portfolio/keramika
│   ├── /portfolio/suvenir
│   ├── /portfolio/kulolchilik
│   └── /portfolio/buyurtma
├── /products (Mahsulotlar va narxlar)
│   ├── /products?category=lepka
│   ├── /products?category=keramika
│   ├── /products?category=suvenir
│   └── /products?category=kulolchilik
├── /certificates (Sertifikatlar va yutuqlar)
├── /about (Biz haqimizda)
└── /contact (Murojaat)
```

3.2 Navigatsiya

Desktop: gorizontal menyu, o'ng tomonda "Murojaat qoldirish" CTA tugmasi
Mobile: hamburger menyu, to'liq ekran overlay menyu

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4. Frontend — Sahifalar spetsifikatsiyasi

4.1 Bosh sahifa (/)

Hero qism

- **To'liq ekran** (`100vh`) vizual effekt
- **Fon:** Usta ishini ko'rsatuvchi yuqori sifatli rasm yoki video
- **Sarlavha:** "Qo'l mehnati bilan yaratilgan go'zallik"
- **Kichik sarlavha:** "Lepka, keramika va kulolchilik ustaxonasi — X&X Ceramics"
- **CTA tugmalar:** "Ishlarimizni ko'rish" → `/portfolio` va "Buyurtma berish" → `/contact`
- **Scroll indikatori:** Pastga o'q animatsiyasi

Xizmatlar qismi

- 6 ta kartochka: Lepka, Tyaga, Keramika, Suvenir haykalchalar, Kulolchilik, Buyurtma asosida
- Har bir kartochkada: ikon, nom, qisqa tavsif

- Hover effekti: ko'tarish animatsiyasi

Tanlangan ishlar (Portfolio preview)

- 3–4 ta eng yaxshi ish masonry grid da
- "Barcha ishlarni ko'rish" tugmasi

Statistika qismi

- Raqamlar animatsiyasi bilan: Yillar tajribasi, Bajarilgan buyurtmalar, Mamnun mijozlar, Sertifikatlar soni

Mijozlar fikrlari (Testimonials)

- Slider/carousel formatida

CTA qism

- "Buyurtma berish" to'liq ekranli qism

4.2 Ishlar portfeli (/portfolio)

Filtr tizimi

```
[ Hammasi ] [ Lepka ] [ Tyaga ] [ Keramika ] [ Suvenir ] [ Kulolchilik ] [ Buyurtma ]
```

- URL query parameter orqali: `/portfolio?category=lepka`
- Framer Motion bilan filtrlash animatsiyasi

Grid formatlar

- Desktop: 3 ustunli masonry grid
- Tablet: 2 ustunli
- Mobile: 1 ustunli

Har bir ish kartochkasi

- Rasm (to'ldiruvchi)
- Nomi
- Kategoriya tegi
- Qisqa tavsif (ixtiyoriy)
- Click → modal yoki alohida sahifa (lightbox)

Lightbox (modal)

- Katta rasm + yon taraflarda navigatsiya o'qlari
- Rasm galereyasi (bir loyiha uchun bir necha rasm)
- Quyida: sarlavha, tavsif, materiallar, o'lcham
- "Shunga o'xshash buyurtma berish" tugmasi

4.3 Mahsulotlar va narxlar (/products)

> **Muhim:** Faqat tayyor va standart mahsulotlar narx bilan ko'rsatiladi. Buyurtma asosidagi mahsulotlar narx bilan ko'rsatilmaydi — ular portfolio ga yo'naltiriladi.

Kategoriya filtrlari (tabs)

```
[ Hammasi ] [ Lepka ] [ Keramika ] [ Suvenir haykalchalar ] [ Kulolchilik ]
```

Mahsulot kartochkasi

```
┌──────────────────┐
│    [Rasm]        │
│                  │
├──────────────────┤
│ Mahsulot nomi    │
│ Kategoriya tegi  │
│                  │
│ Tavsif (qisqa)   │
│                  │
│    150,000 UZS   │
│  [Murojaat →]    │
└──────────────────┘
```

Maydonlar:

- Rasm (1 yoki ko'p)
- Mahsulot nomi (uz/ru)
- Kategoriya
- Qisqa tavsif
- Narx (UZS)
- O'lchamlar (ixtiyoriy)
- Mavjudligi: "Mavjud" / "Buyurtma asosida" / "Tugagan"
- "Murojaat qoldirish" tugmasi (Telegram yoki forma)

4.4 Sertifikatlar va yutuqlar (/certificates)

Yutuqlar timeline

- Vertikal timeline formatida
- Har bir yutuq: yil, nomi, tavsif, rasm (ixtiyoriy)

Sertifikatlar galereyasi

- Kartochka formatida sertifikat rasmlari
- Modal da katta ko'rish imkoni

Ko'rsatkichlar qismi

- Tanlovlarda ishtirok, sovrinlar, e'tiroflar

4.5 Biz haqimizda (/about)

- Korxona tarixi
- Xasanboy va Xusanboy haqida (foto bilan)
- Ustaxona haqida (ishlab chiqarish jarayoni)
- Qadriyatlar
- Hamkorlar (bo'lsa)

4.6 Murojaat (/contact)

Murojaat formasi

```
Ism va familiya *
Telefon raqam *
E-mail
Murojaat turi: [ Mahsulot sotib olish ] [ Buyurtma berish ] [ Hamkorlik ] [ Boshqa ]
Qo'shimcha ma'lumot (textarea)
Rasm yuklash (ixtiyoriy, max 5 ta, max 10MB har biri)
[ Yuborish ]
```

Murojaat yo'llari

- Telegram: @xxceramics
- Telefon: +998 XX XXX XX XX
- Instagram: @xx_ceramics

- Manzil: [Shahar, ko'cha]
- Ish vaqti: Du-Sha, 09:00–18:00

Google Maps

- Ustaxona joylashuvi (embed)

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5. Frontend — Dizayn tizimi

5.1 Rang palitasi

```css
:root {
  /* Asosiy ranglar */
  --color-primary: #8B5E3C;       /* Issiq jigarrang (keramika tusi) */
  --color-primary-light: #C4986B; /* Och jigarrang */
  --color-primary-dark: #5C3D22;  /* To'q jigarrang */

  /* Fon ranglar */
  --color-bg: #FAF7F2;            /* Kremsimon oq */
  --color-bg-secondary: #F0EBE3;  /* Och beige */
  --color-surface: #FFFFFF;       /* Toza oq */

  /* Matn ranglar */
  --color-text: #2C1810;          /* To'q jigarrang */
  --color-text-secondary: #6B4A35; /* O'rta jigarrang */
  --color-text-muted: #A08070;    /* Och matn */

  /* Aksent */
  --color-accent: #D4843C;        /* To'q sariq-jigarrang */
  --color-accent-green: #5C7A4E;  /* O'rmon yashil */

  /* Chiziqlar */
  --color-border: #E0D5C8;
}
```

5.2 Tipografiya

```css
```

```
/* Sarlavha uchun */
font-family: 'Playfair Display', Georgia, serif;

/* Asosiy matn uchun */
font-family: 'Inter', system-ui, sans-serif;

/* O'lchov shkalasi */
--text-xs: 0.75rem;      /* 12px */
--text-sm: 0.875rem;     /* 14px */
--text-base: 1rem;       /* 16px */
--text-lg: 1.125rem;     /* 18px */
--text-xl: 1.25rem;      /* 20px */
--text-2xl: 1.5rem;      /* 24px */
--text-3xl: 1.875rem;    /* 30px */
--text-4xl: 2.25rem;     /* 36px */
--text-5xl: 3rem;        /* 48px */
--text-6xl: 3.75rem;     /* 60px */
```

5.3 Komponent kutubxonasi

Tugmalar (Buttons)

```
Primary:   To'q jigarrang fon, oq matn, hover da 10% to'qroq
Secondary: Shaffof fon, jigarrang chegara, hover da och fon
Ghost:     Fonsiz, matn rangida, hover da pastki chiziq
```

Kartochkalar (Cards)

```
Default:  Oq fon, 1px chegara, 12px radius, 24px padding
Hover:    Ko'tarish effekti (box-shadow kuchayishi)
Featured: Birlamchi rang chegara
```

Formalar

```
Input:     Kremsimon fon, 1px chegara, focus da jigarrang outline
Textarea:  Kengayuvchi, min 120px balandlik
Select:    Custom o'q, jigarrang rang
```

5.4 Animatsiyalar

javascript

```
// Sahifaga kirish animatsiyasi
const fadeInUp = {
  initial: { opacity: 0, y: 30 },
  animate: { opacity: 1, y: 0 },
  transition: { duration: 0.6, ease: "easeOut" }
};

// Scroll trigger (Intersection Observer)
// Raqamlar counter animatsiyasi: 0 dan target gacha, 2 soniyada
// Filtr o'tish: layout animatsiya (Framer Motion AnimatePresence)
// Hover effektlari: CSS transition, 200ms
```

5.5 Responsive breakpoints

```css
sm:  640px   /* Mobile landscape */
md:  768px   /* Tablet */
lg:  1024px  /* Kichik desktop */
xl:  1280px  /* Desktop */
2xl: 1536px  /* Katta ekran */
```

5.6 SEO va meta

Har bir sahifa uchun:

```html
<title>X&X Ceramics | Sahifa nomi</title>
<meta name="description" content="..."/>
<meta property="og:image" content="..."/>
<!-- Structured data (JSON-LD): LocalBusiness -->
<!-- Sitemap.xml avtomatik generatsiya -->
<!-- robots.txt -->
```

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6. Backend — API spetsifikatsiyasi

6.1 Asosiy URL tuzilishi

```
Base URL: https://api.xxceramics.uz/v1
```

6.2 Mahsulotlar API

```
GET    /products                Barcha mahsulotlar (filtrlash bilan)
```

```
GET    /products/:id              Bitta mahsulot
POST   /products                  Yangi mahsulot (Admin)
PUT    /products/:id              Tahrirlash (Admin)
DELETE /products/:id              O'chirish (Admin)
POST   /products/:id/images       Rasm yuklash (Admin)
DELETE /products/:id/images/:imgId Rasm o'chirish (Admin)
```

GET /products — query parametrlar:

```
?category=lepka|keramika|suvenir|kulolchilik
?available=true|false
?page=1&limit=12
?sort=price_asc|price_desc|created_desc
```

Mahsulot ob'ekti:

json

```
{
  "id": "uuid",
  "name_uz": "Lepka idish",
  "name_ru": "Лепка тарелка",
  "description_uz": "...",
  "description_ru": "...",
  "category": "lepka",
  "price": 150000,
  "currency": "UZS",
  "images": ["url1", "url2"],
  "dimensions": "20x15x10 sm",
  "weight": "500g",
  "material": "Loy, sirli bo'yoq",
  "availability": "available|order|sold_out",
  "is_custom_order": false,
  "created_at": "2025-01-01T00:00:00Z",
  "updated_at": "2025-01-01T00:00:00Z"
}
```

6.3 Portfolio API

```
GET    /portfolio                 Barcha ishlar
GET    /portfolio/:id             Bitta ish
POST   /portfolio                 Yangi ish (Admin)
PUT    /portfolio/:id             Tahrirlash (Admin)
DELETE /portfolio/:id             O'chirish (Admin)
```

Portfolio ob'ekti:

json

```
{
  "id": "uuid",
  "title_uz": "Gilam naqshli ko'za",
  "title_ru": "Ваза с орнаментом",
  "category": "kulolchilik",
  "description_uz": "...",
  "images": ["url1", "url2", "url3"],
  "year": 2024,
  "is_featured": true,
  "tags": ["ko'za", "naqsh", "buyurtma"],
  "created_at": "..."
}
```

6.4 Murojaatlar API

```
POST   /contacts                Murojaat yuborish (Public)
GET    /contacts                Barcha murojaatlar (Admin)
GET    /contacts/:id            Bitta murojaat (Admin)
PATCH  /contacts/:id/status     Holat o'zgartirish (Admin)
```

Murojaat ob'ekti:

json

```
{
  "id": "uuid",
  "name": "Alisher Karimov",
  "phone": "+998901234567",
  "email": "alisher@mail.com",
  "type": "order|purchase|partnership|other",
  "message": "...",
  "attachments": ["url1"],
  "status": "new|viewed|processing|completed|cancelled",
  "created_at": "..."
}
```

6.5 Sertifikatlar API

```
GET    /certificates            Barcha sertifikatlar
POST   /certificates            Qo'shish (Admin)
PUT    /certificates/:id        Tahrirlash (Admin)
DELETE /certificates/:id        O'chirish (Admin)
```

6.6 Auth API

```
POST   /auth/login              Admin kirish
POST   /auth/refresh            Token yangilash
POST   /auth/logout             Chiqish
```

```json
// POST /auth/login request
{ "username": "admin", "password": "..." }

// Response
{
  "access_token": "jwt...",
  "refresh_token": "jwt...",
  "expires_in": 3600
}
```

6.7 Xavfsizlik

- **CORS:** Faqat xxceramics.uz domeniga ruxsat
- **Rate limiting:** 100 so'rov/daqiqa (public), 500 (admin)
- **Helmet.js:** HTTP xavfsizlik sarlavhalari
- **Input sanitization:** XSS himoyasi
- **File upload:** Faqat rasm formatlari (jpg, jpeg, png, webp, gif), max 10MB
- **JWT:** Access token 1 soat, Refresh token 7 kun

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7. Ma'lumotlar bazasi sxemasi

7.1 Asosiy jadvallar

```sql
```

```sql
-- Foydalanuvchilar (admin)
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  username VARCHAR(50) UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  role VARCHAR(20) DEFAULT 'admin',
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Mahsulotlar
CREATE TABLE products (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name_uz VARCHAR(200) NOT NULL,
  name_ru VARCHAR(200),
  description_uz TEXT,
  description_ru TEXT,
  category VARCHAR(50) NOT NULL,
  price INTEGER,  -- UZS da, null bo'lsa buyurtma asosida
  currency VARCHAR(10) DEFAULT 'UZS',
  availability VARCHAR(20) DEFAULT 'available',
  is_custom_order BOOLEAN DEFAULT false,
  dimensions VARCHAR(100),
  weight VARCHAR(50),
  material VARCHAR(200),
  is_active BOOLEAN DEFAULT true,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Mahsulot rasmlari
CREATE TABLE product_images (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  product_id UUID REFERENCES products(id) ON DELETE CASCADE,
  url TEXT NOT NULL,
  cloudinary_id TEXT,
  is_primary BOOLEAN DEFAULT false,
  sort_order INTEGER DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Portfolio
CREATE TABLE portfolio_items (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title_uz VARCHAR(200) NOT NULL,
  title_ru VARCHAR(200),
  category VARCHAR(50) NOT NULL,
```

```
  description_uz TEXT,
  description_ru TEXT,
  year INTEGER,
  is_featured BOOLEAN DEFAULT false,
  is_active BOOLEAN DEFAULT true,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Portfolio rasmlari
CREATE TABLE portfolio_images (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  portfolio_id UUID REFERENCES portfolio_items(id) ON DELETE CASCADE,
  url TEXT NOT NULL,
  cloudinary_id TEXT,
  is_primary BOOLEAN DEFAULT false,
  sort_order INTEGER DEFAULT 0
);

-- Murojaatlar
CREATE TABLE contacts (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(200) NOT NULL,
  phone VARCHAR(20) NOT NULL,
  email VARCHAR(200),
  type VARCHAR(50) NOT NULL,
  message TEXT,
  status VARCHAR(20) DEFAULT 'new',
  telegram_notified BOOLEAN DEFAULT false,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Murojaat rasmlari
CREATE TABLE contact_attachments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  contact_id UUID REFERENCES contacts(id) ON DELETE CASCADE,
  url TEXT NOT NULL,
  cloudinary_id TEXT
);

-- Sertifikatlar
CREATE TABLE certificates (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title_uz VARCHAR(300) NOT NULL,
  title_ru VARCHAR(300),
  description_uz TEXT,
```

```
  description_ru TEXT,
  year INTEGER,
  issued_by VARCHAR(200),
  image_url TEXT,
  cloudinary_id TEXT,
  type VARCHAR(50) DEFAULT 'certificate',  -- certificate|award|achievement
  is_active BOOLEAN DEFAULT true,
  sort_order INTEGER DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

---

8. Telegram Bot — Admin Panel

8.1 Bot arxitekturasi

- **Kutubxona:** `node-telegram-bot-api` yoki `grammy`
- **Webhook:** HTTPS orqali real vaqtda bildirishnomalar
- **Admin identifikatsiya:** Telegram `chat_id` ro'yxati (config da saqlash)

8.2 Bot buyruqlari

```
/start        — Bot haqida ma'lumot va menyu
/help         — Buyruqlar ro'yxati
/products     — Mahsulotlar boshqaruvi menyusi
/add_product  — Yangi mahsulot qo'shish
/contacts     — Murojaatlar ro'yxati
/stats        — Statistika
```

8.3 Yangi mahsulot qo'shish jarayoni

Bot foydalanuvchini qadamma-qadam so'rovlar orqali boshqaradi:

```
1. /add_product
   → "Mahsulot kategoriyasini tanlang:"
   [Lepka] [Tyaga] [Keramika]
   [Suvenir] [Kulolchilik]

2. Kategoriya tanlangandan so'ng:
   → "Mahsulot nomini kiriting (o'zbek tilida):"
   ← [Foydalanuvchi nomi yozadi]

3. → "Rus tilida nomi (ixtiyoriy, o'tkazib yuborish uchun -):"
   ← [Nom yoki -]

4. → "Tavsif kiriting (o'zbek tilida):"
   ← [Tavsif]

5. → "Narxni kiriting (UZS da, buyurtma asosida bo'lsa 0 kiriting):"
   ← [150000 yoki 0]

6. → "Mavjudligini tanlang:"
   [Mavjud] [Buyurtma asosida] [Tugagan]

7. → "Rasmlarni yuboring (max 5 ta). Tugagach 'Tayyor' bosing:"
   ← [Rasmlar yuboriladi]
   ← [Tayyor tugmasi]

8. → "Ma'lumotlarni tekshiring:"
   [Preview ko'rinishi]
   [✅ Saqlash] [❌ Bekor qilish]

9. → "✅ Mahsulot muvaffaqiyatli qo'shildi!"
```

8.4 Narx o'zgartirish

```
/products →
  [Ro'yxat, pagination bilan]
  Mahsulot tanlash →
  [✏️ Narxni o'zgartirish] [📝 Tahrirlash] [🗑️ O'chirish]

Narx o'zgartirish:
  → "Yangi narxni kiriting (UZS):"
  ← [Yangi narx]
  → "✅ Narx yangilandi: 150,000 → 200,000 UZS"
```

8.5 Murojaatlar bildirishnomasi

Yangi murojaat kelganda bot adminlarga darhol xabar yuboradi:

```
🔔 Yangi murojaat!

👤 Ism: Alisher Karimov
📞 Telefon: +998 90 123 45 67
📧 Email: alisher@mail.com
📂 Turi: Buyurtma berish
💬 Xabar: "Toychoq uchun katta ko'za buyurtma qilmoqchiman..."

[📞 Qo'ng'iroq qilish] [✅ Ko'rildi] [➡️ Jarayonda]
```

8.6 Statistika

```
/stats →
📊 X&X Ceramics statistikasi

📦 Jami mahsulotlar: 45
🆕 Bu oy qo'shilgan: 3
📨 Jami murojaatlar: 128
🆕 Yangi (ko'rilmagan): 5
✅ Yakunlangan: 98

[📅 Bu hafta] [📅 Bu oy] [📅 Umumiy]
```

---

9. Fayl saqlash — Cloudinary

9.1 Papkalar tuzilishi

```
xxceramics/
├── products/          # Mahsulot rasmlari
├── portfolio/         # Portfolio rasmlari
├── certificates/      # Sertifikat rasmlari
└── contacts/          # Murojaat rasmlari (vaqtinchalik)
```

9.2 Rasm optimizatsiyasi

javascript

```
// Cloudinary transformatsiyalar
const productImage = {
  width: 800,
  height: 800,
  crop: 'fill',
  gravity: 'center',
  quality: 'auto:good',
  format: 'webp'
};

const thumbnail = {
  width: 400,
  height: 400,
  crop: 'fill',
  quality: 'auto:eco',
  format: 'webp'
};
```

---

10. Hosting va Deploy

10.1 Tavsiya etilgan hosting

Xizmat	Maqsad	Narxi (taxminiy)
Vercel	Frontend (Next.js)	Bepul (Hobby) / $20/oy (Pro)
Railway.app yoki Render	Backend (Node.js)	$5–20/oy
Supabase	PostgreSQL	Bepul / $25/oy
Cloudinary	Rasm saqlash	Bepul (25GB)

10.2 Domen

- **Asosiy:** xxceramics.uz (O'zbekiston domeniga afzallik)
- **Zaxira:** xxceramics.com
- **SSL:** Let's Encrypt (bepul)

10.3 CI/CD

```
GitHub → GitHub Actions → Vercel (frontend)
                        → Railway (backend)
```

11. Ko'p tillilik (i18n)

11.1 Qo'llab-quvvatlanadigan tillar

- **O'zbek (UZ)** — asosiy til
- **Rus (RU)** — ikkinchi til

11.2 Amalga oshirish

```javascript
// next-intl kutubxona ishlatish
// Matn fayllari: /messages/uz.json va /messages/ru.json

// URL tuzilishi:
xxceramics.uz/uz/...  (o'zbek)
xxceramics.uz/ru/...  (rus)
```

---

12. Ishlash samaradorligi (Performance)

12.1 Maqsad ko'rsatkichlar

Ko'rsatkich	Maqsad
LCP (Largest Contentful Paint)	< 2.5 soniya
FID (First Input Delay)	< 100 ms
CLS (Cumulative Layout Shift)	< 0.1
Lighthouse Score	> 90
Time to First Byte (TTFB)	< 800 ms

12.2 Optimizatsiya usullari

- Rasmlar: WebP format, lazy loading, `next/image`
- Kod: Code splitting, dynamic import
- Cache: Statik sahifalar CDN da cache
- Font: `font-display: swap`, Google Fonts local hosting
- API: Pagination, ma'lumotlarni cache'lash (Redis yoki in-memory)

---

13. Loyihani amalga oshirish rejasi

13.1 Fazalar

Faza 1 (2–3 hafta): Poydevor

- Loyiha strukturasini sozlash (Next.js, Express, PostgreSQL)
- Ma'lumotlar bazasi sxemasini yaratish
- Asosiy API endpointlarini yozish (mahsulotlar, portfolio, murojaatlar)
- Auth tizimini sozlash

Faza 2 (2–3 hafta): Frontend asosiy sahifalar

- Dizayn tizimini yaratish (rang, tipografiya, komponentlar)
- Bosh sahifa, Portfolio, Mahsulotlar sahifalarini yaratish
- Murojaat formasi
- Responsive dizayn

Faza 3 (1–2 hafta): Qo'shimcha sahifalar va Telegram bot

- Sertifikatlar va yutuqlar sahifasi
- Biz haqimizda sahifasi
- Telegram bot yaratish va ulash
- Bildirishnoma tizimi

Faza 4 (1 hafta): Sinovlar va deploy

- Cross-browser testing
- Mobile testing
- SEO optimizatsiyasi
- Production deploy
- Ma'lumotlarni boshlang'ich kiritish

Jami taxminiy muddat: 6–9 hafta

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14. Xavfsizlik talablari

- **HTTPS** barcha aloqalar uchun majburiy
- **Admin panel** faqat Telegram bot orqali — alohida web admin panel yo'q
- **JWT token** saqlash: `httpOnly cookie` (localStorage emas)
- **Rasm yuklash** validatsiyasi: fayl turi va hajmi tekshirish
- **SQL injection** himoyasi: Prisma ORM ishlatish

- **Bot admin** ro'yxati: faqat belgilangan `chat_id` lar

---

15. Test talablari

15.1 Frontend

- **Unit test:** React komponentlar (Jest + React Testing Library)
- **E2E test:** Murojaat formasi, filtrlar (Playwright)

15.2 Backend

- **Unit test:** API endpointlar (Jest + Supertest)
- **Integration test:** Ma'lumotlar bazasi operatsiyalari

15.3 Qo'lda test

- Barcha brauzerlar: Chrome, Firefox, Safari, Edge
- Mobile: iOS Safari, Android Chrome
- Sekin internet (3G simulyatsiya)

Ilovalar

A: Kategoriyalar ro'yxati

Kod	Nomi (UZ)	Nomi (RU)
`lepka`	Lepka	Лепка
`tyaga`	Tyaga	Тяга
`keramika`	Keramika	Керамика
`suvenir`	Suvenir haykalchalar	Сувенирные статуэтки
`kulolchilik`	Kulolchilik mahsulotlari	Гончарные изделия
`buyurtma`	Buyurtma asosida	На заказ

B: Murojaat turlari

Kod	Nomi
`purchase`	Tayyor mahsulot sotib olish

Kod	Nomi
`order`	Buyurtma asosida yasatish
`partnership`	Hamkorlik taklifi
`other`	Boshqa

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